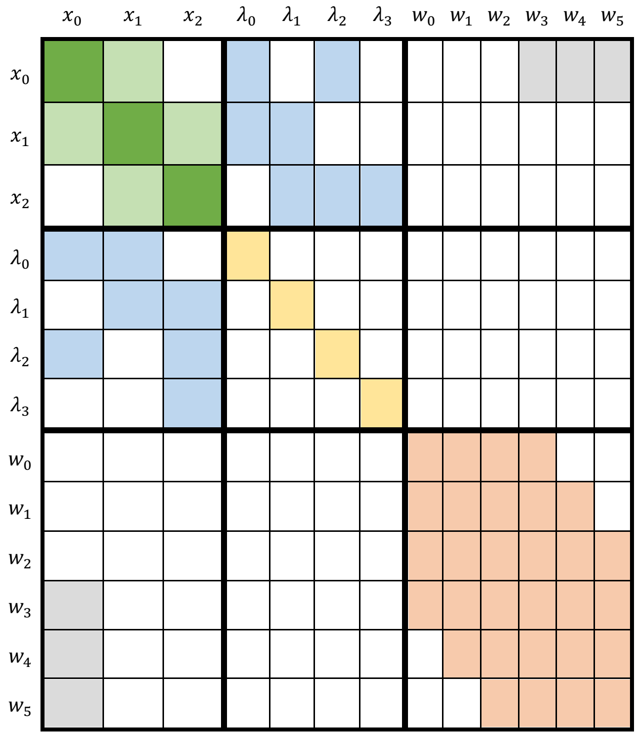
The Definition of the Information Matrix for the UWB Factor

The information matrix for the UWB factor in the manuscript is illustrated in the figure below. For clarity, the short sliding window states **x** consist of 3 keyframes (x_0, x_1, x_2), the visual reprojections $\boldsymbol{\lambda}$ consist of 4 feature points $(\lambda_0, \lambda_1, \lambda_2, \lambda_3)$, and the long sliding window states **w** consist of 6 key trajectory points $(w_0, w_1, w_2, w_3, w_4, w_5)$.



As shown in the figure, this information matrix has a sparsity pattern similar to that of conventional VIO. On this basis, the UWB factor (Eq. (5)) introduces additional structural variations, mainly reflected in the coupling between the long and cross sliding windows, as follows:

1) The long-window ranging factor R_l acts on key trajectory points at the same time, forming diagonal constraints in the (w, w) sub-block (orange part); combined with pose graph (PG) constraints, it also forms a banded sparse structure near the main diagonal.

2) The short-window ranging factor R_s only affects the short sliding window states and primarily enhances the diagonal information of the position components (**dark green** part in the (x, x) sub-block), which does not alter the overall sparsity pattern.

3) Cross-window coupling is only introduced by pose graph constraints, corresponding to the (x, w) sub-block (gray part), with non-zero entries only between the oldest short-window keyframe x_0 and the most recent long-window key trajectory points (e.g., w_3, w_4, w_5).

4) Since the UWB residuals only relate to positional information and do not involve visual feature parameters, the remaining (λ, w) sub-blocks are all zero.